

UNIT - 2

Function of complex function

A complex variable is denoted by $z = x + iy$ where x and y are real variables

Functions of a complex variable:

$w = u(x, y) + iv(x, y)$ is a function of the complex variable $z = x + iy$ (i.e.) $w = f(z) = u(x, y) + iv(x, y)$

where $u(x, y)$ is the real part of $f(z)$ and $v(x, y)$ is the imaginary part of the complex function $f(z)$

In general we can write $f(z)$ as $f(z) = u + iv$

Limit of a function:

Let $f(z)$ be a function defined in a set S and let z_0 be a limit point of S then A is said to be limit point of $f(z)$ at z_0

if for any $\epsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - A| < \epsilon$ for all z in S other than z_0 with $|z - z_0| < \delta$ it is denoted by

$$\lim_{z \rightarrow z_0} f(z) = A, \quad |z - z_0| < \delta$$

Continuity of a function:-

Let $f(z)$ be a function defined in S and let z_0 be a limit point if the limit of $f(z)$ at z_0 exists and if it is finite and is equal to

$$f(z_0) \quad \lim_{z \rightarrow z_0} f(z) = f(z_0)$$

then $f(z)$ is said to be continuous at

z_0 Derivative of a complex function:-

A function w equal to $f(z)$ is said to be differentiable at point z_0 is equal to z_0 if the

limit $\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ exists,

And it is denoted by $f'(z_0)$

$$(i.e) f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Analytic function (or) regular function.
(or) holomorphic function.

A function defined at a point z_0 is said to be analytic at z_0 if it has a derivative at z_0 and at every point in some neighbourhood of z_0 . It is said to be analytic in region R if it is analytic at every point of R .

Singular point:

A point at which the function $w = f(z)$ fails to be analytic is called a singular point or singularity of $f(z)$.

State & Prove
Necessity condition for $f(z)$ to be
analytic (or) Cauchy - Riemann equation in
cartesian coordinates. Unit - I

Statement : (i) necessity condition:

If $w = f(z) = u + iv$ is an
analytic function of z then the
four partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$
should exist and
satisfy $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

(i.e) $u_x = v_y$ & $v_x = -u_y$

Proof:

Let $f(z) = u(x, y) + iv(x, y)$
be an analytic function at the
point z in a region R
since $f(z)$ is analytic its derivative
 $f'(z)$ exists in R and is
given by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{--- (1)}$$

$$\text{Let } z = x + iy$$

$$\Delta z = \Delta x + i\Delta y$$

$$z + \Delta z = (x + iy) + (\Delta x + i\Delta y)$$

$$z + \Delta z = x + \Delta x + i(y + \Delta y)$$

$$f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y)$$

$$\textcircled{1} \Rightarrow f'(z) = \lim_{\Delta z \rightarrow 0} \frac{[u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y)] - [u(x, y) + i v(x, y)]}{\Delta x + i\Delta y}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{\{u(x + \Delta x, y + \Delta y) - u(x, y)\} + i\{v(x + \Delta x, y + \Delta y) - v(x, y)\}}{\Delta x + i\Delta y} \quad - \textcircled{2}$$

Now let us assume that

(i) Δx tends to zero and $\Delta y = 0$

$$f'(z) = \lim_{\Delta x \rightarrow 0} \frac{\{u(x + \Delta x, y) - u(x, y)\} + i\{v(x + \Delta x, y) - v(x, y)\}}{\Delta x}$$

$$f'(z) = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}$$

$$f'(z) = u_x + i v_x \quad \text{--- (3)}$$

Next we assume that

(ii) $\Delta x = 0$ and Δy tends to zero.

then equation (2) becomes,

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{\{u(x, y+\Delta y) - u(x, y)\} + i\{v(x, y+\Delta y) - v(x, y)\}}{i \Delta y}$$

multiply & divide by conjugate.

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{\{u(x, y+\Delta y) - u(x, y)\} + i\{v(x, y+\Delta y) - v(x, y)\}}{\Delta y} \times \frac{-i}{-i}$$

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{-i\{u(x+\Delta x, y) - u(x, y)\}}{\Delta y} + \lim_{\Delta y \rightarrow 0} \frac{v(x+\Delta x, y) - v(x, y)}{\Delta y}$$

$$f'(z) = -i u_y + v_y \quad \text{--- (4)}$$

From eq (3) & (4), we get

$$u_x + i v_x = -i u_y + v_y$$

Equating real and imaginary parts
on both sides, we get

$$u_x = u_y$$

$$\& \quad v_x = -u_y$$

Hence CR's equation is verified.

Sufficient condition for analytic function

A single value function $w=f(z)$ possessing continuous partial derivatives

$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ at each pt of region

are and satisfying CR equations

then $f(z)$ is analytic in region.

Polar form of Cauchy Riemann equation :-

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

Harmonic function:

Any function of (x) & (y) which possesses continuous first & second order partial derivatives and satisfied Laplace equation is called a harmonic function.

If $f(z) = u + iv$ is an analytic function
 then u and v are harmonic
 function.

$$u_{xx} + u_{yy} = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$v_{xx} + v_{yy} = 0$$

Conjugate harmonic functions:-

If $f(z)$ is equal to $u + iv$
 is analytic function of z then u and
 v are harmonic functions. Such u and
 v are said to be conjugate harmonic
 function.

When v is called the harmonic
 conjugate of u .
 When u is called the harmonic
 conjugate of v .

Properties of analytic function:-

1) If $f(z) = u + iv$ is an analytic
 function of $z = x + iy$ then u & v
 satisfies Laplace equation.

$$\nabla^2 \phi = 0$$

$$(i.e) \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Proof:

given $f(z) = u + iv$ is an analytic function
 $\therefore$ CR are satisfied

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{--- (1)}$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \text{--- (2)}$$

Diff eqn (1) w.r.t x .

Diff eqn (2) w.r.t y .

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{--- (3)}$$

$$\frac{\partial^2 v}{\partial y \partial x} = -\frac{\partial^2 u}{\partial y^2} \rightarrow \text{--- (4)}$$

(3) + (4)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y \partial x} = \frac{\partial^2 v}{\partial x \partial y} + \left(-\frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y \partial x} - \frac{\partial^2 v}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore u$ satisfies Laplace equation.

Diff eqn (1) wrt y ,

Diff eqn (2) wrt x ,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial y^2} \quad - (5)$$

$$\frac{\partial^2 v}{\partial x^2} = -\frac{\partial^2 u}{\partial x \partial y} \quad - (6)$$

(5) + (6)

$$\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 u}{\partial y \partial x}$$

$$\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 u}{\partial y \partial x} = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$\therefore v$ satisfies Laplace equation

(2) if $f(z) = u(x, y) + i v(x, y)$ is an analytic function then the curves $u(x, y) = c_1$ and $v(x, y) = c_2$

are orthogonal to each other,
where c_1 and c_2 are constants.

$$u(x, y) = c_1$$

$$du = 0$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$0 = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$-\frac{\partial u}{\partial x} dx = \frac{\partial u}{\partial y} dy$$

$$-\frac{u_x}{u_y} = \frac{dy}{dx} = m_1$$

$$v(x, y) = c_2$$

$$dv = 0$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

$$0 = v_x dx + v_y dy$$

$$-v_x dx = v_y dy$$

$$-\frac{v_x}{v_y} = \frac{dy}{dx} = m_2$$

$$m_1 m_2 = \frac{-u_x}{u_y} \times \frac{-v_x}{v_y}$$

$$= \frac{-u_x}{u_y} \times \frac{v_y}{v_x}$$

$$m_1 m_2 = -1$$

if $f(z)$ is analytic in region R and $f'(z) = 0$ everywhere in R show that $f(z)$ is constant.

$f(z) = u + iv$ is an analytic function and $\therefore f'(z) = 0$

diff w.r.t z :

$$f'(z) = u_x + i v_x$$

$$0 = u_x + i v_x$$

$$0 + i0 = u_x + i v_x$$

$$u_x = 0 \quad \& \quad v_x = 0$$

gives $f(z)$ is analytic $\therefore$ it satisfies CR equations,

$$u_x = v_y$$

$$v_x = -u_y$$

$$u_x = v_y$$

$$v_x = -u_y$$

$$u_x = 0 \rightarrow v_y = 0$$

$$v_x = 0 \Rightarrow -u_y = 0$$

$$\Rightarrow u_y = 0$$

$$u_x = 0 \text{ \& } u_y = 0 \Rightarrow u = c$$

$$u_x = 0 \text{ \& } v_y = 0 \Rightarrow v = c$$

$$f(z) = u + iv$$

$$= c + ic$$

$$f(z) = c$$

P.T that given f is an analytic fn with constant real part is constant

given $f(z) = u + iv$ is an analytic function and $u = \text{constant}$

$$u_x = 0$$

$$u_y = 0$$

$\therefore$ it satisfies CR equation

$$u_x = v_y$$

$$v_x = -u_y$$

$$u_x = 0 \Rightarrow v_y = 0$$

$$u_y = 0 \Rightarrow v_x = 0$$

$$v_x = 0 \quad \& \quad v_y = 0$$

$$v = c$$

$$f(z) = u + iv \\ = c + ic$$

$$f(z) = c$$

P.T an analytic fn with constant modulus is constant.

function gives $f(z) = u + iv$ is an analytic also gives $|f(z)| = c$.

$$|u + iv| = c$$

$$\sqrt{u^2 + v^2} = c$$

$$u^2 + v^2 = c^2 \quad \text{--- (1)}$$

Diff eqn (1) w.r.t x & y

$$2u u_x + 2v v_x = 0$$

$$2u u_y + 2v v_y = 0$$

$$u u_x + v v_x = 0 \rightarrow (2)$$

$$u u_y + v v_y = 0 \rightarrow (3)$$

$\downarrow$ $\downarrow$
 v_x u_y

$$(3) \Rightarrow -u v_x + v u_x = 0 \rightarrow (4)$$

Taking eq (2) & (4)

$$u u_x + v v_x = 0 \quad - \quad (2)$$

$$-u v_x + v u_x = 0 \rightarrow (4)$$

$$(2) \times u \Rightarrow u^2 u_x + u v v_x = 0$$

$$(4) \times v \Rightarrow v^2 u_x - u v v_x = 0$$

$$u^2 u_x + v^2 u_x = 0$$

$$u_x (u^2 + v^2) = 0$$

$$u_x (u^2 + v^2) = 0$$

$$u_x = 0$$

Sub $u_x = 0$ in eq (2)

$$v_x = 0$$

$$u_x = v_y$$

$$v_x = -u_y$$

$$u_x = 0 \Rightarrow v_y = 0$$

$$v_x = 0 \Rightarrow u_y = 0$$

$$u_x = 0 \text{ \& } u_y = 0 \rightarrow u = C$$

$$v_x = 0 \text{ \& } v_y = 0 \rightarrow v = C$$

$$f(z) = u + iv$$

$$f(z) = C + iC$$

$$f(z) = C$$

Q. In $f(z)$ & $\overline{f(z)}$ are analytic function then $f(z)$ is constant

given $f(z) = u + iv$ is an analytic function.

$\therefore$ it satisfies CR equations
(i.e.)
$$\left. \begin{aligned} u_x &= v_y \\ v_x &= -u_y \end{aligned} \right\} \text{--- (1)}$$

Also given $\overline{f(z)} = \overline{u + iv}$

also given $\overline{f(z)} = u - iv$
 $\therefore$ it satisfies CR equations

$$\left. \begin{aligned} u_x &= -v_y \\ v_x &= u_y \end{aligned} \right\} \text{--- (2)}$$

Add eq (1) + (2)

$$\rightarrow 2u_x = 0$$

Add eq (1) + (2)

$$\rightarrow 2v_x = 0$$

$$\rightarrow u_x = 0$$

$$v_x = 0$$

$$u_x = 0 \rightarrow u_y = 0$$

$$v_x = 0 \rightarrow v_y = 0$$

$$u_x = 0 + u_y = 0 \rightarrow u = C$$

$$v_x = 0 + v_y = 0 \rightarrow v = C$$

$$f(z) = u + iv$$

$$f(z) = C + iC$$

$$f(z) = C$$

Problems :-

1. Test the analyticity of the function

1b) $w = \sin z$ $w = z^2$

$$w = f(z) = u + iv$$

$$w = \sin z$$

$$z = x + iy$$

$$u + iv = \sin(x + iy)$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y \quad \begin{matrix} \cos i\theta = \cosh \theta \\ \sin i\theta = i \sinh \theta \end{matrix}$$

$$u = \sin x \cosh y$$

$$v = \cos x \sinh y$$

Diff w w.r.t to x & y .

$$u_x = \cos x \cosh y \quad u_y = \sin x \sinh y$$

$$v_x = -\sin x \sinh y \quad v_y = \cos x \cosh y$$

$$u_x = v_y$$

$$v_x = -u_y$$

$\therefore$ gnfn $w = \sin z$ is an analytic function.

gn hyperbolic
funct
+ve
rot
cosh y
sinh y

$$w = \cos z$$

$$w = \cos z$$

$$u + iv = \cos(x + iy)$$

$$u + iv = \cos x \cosh y - i \sin x \sinh y$$

$$u + iv = \cos x \cosh y - i \sin x \sinh y$$

$$u = \cos x \cosh y$$

$$v = -\sin x \sinh y$$

$$u_x = -\sin x \cosh y, \quad u_y = \cos x \sinh y$$

$$v_x = -\cos x \sinh y, \quad v_y = -\sin x \cosh y$$

$$u_x = v_y$$

$$v_x = -u_y$$

$\therefore w = \cos z$ is an analytic function.

will

Verify the analyticity of $w = \cosh z$.

$$w = \cosh z$$

Sol:-

$$w = \cosh z, \quad w = \cosh z$$

$$u + iv = \cosh(x + iy)$$

$$= \cosh(x + i2y)$$

$$u + iv = \cos(i\pi - y)$$

$$u + iv = \cos i\pi \cos y + \sin i\pi \sin y$$

$$u + iv = \cosh x \cos y + i \sinh x \sin y$$

$$u = \cosh x \cos y$$

$$v = \sinh x \sin y$$

$$u_x = \sinh x \cos y, \quad u_y = -\cosh x \sin y$$

$$u_x = \sinh x \cos y$$

$$u_y = -\cosh x \sin y$$

$$v_x = \cosh x \sin y$$

$$v_y = \sinh x \cos y$$

$$u_x = v_y$$

$$v_x = -u_y$$

$w = \cosh z$ is an analytic function

verify the analyticity of the fn

$$w = \sinh z$$

$$\sin iz = i \sinh z$$

$$\frac{1}{i} \sin iz = \sinh z$$

$$\frac{-i}{-i} \times \frac{1}{i} \sin iz = \sinh z$$

$$-i \sin i\pi = \sinh \pi$$

$$-i \sin i(\pi + i\eta) = \sinh \pi$$

$$-i \sin(i\pi - \eta) = \sinh \pi$$

$$w = \sinh \pi$$

$$w = -i \sin(i\pi - \eta)$$

$$u + iv = -i [\sin i\pi \cosh \eta - \cos i\pi \sinh \eta]$$

$$u + iv = -i [\sinh \pi \cosh \eta - \cosh \pi \sinh \eta]$$

$$u + iv = \sinh \pi \cosh \eta + i \cosh \pi \sinh \eta$$

$$u = \sinh \pi \cosh \eta$$

$$v = \cosh \pi \sinh \eta$$

$$u_x = \cosh \pi \cosh \eta$$

$$u_y = -\sinh \pi \sinh \eta$$

$$v_x = \sinh \pi \sinh \eta$$

$$v_y = \cosh \pi \cosh \eta$$

$$u_x = -u_y$$

$$v_x = -v_y$$

$w = \sinh \pi$ is an analytic function

verify for $2xy + i(x^2 - y^2)$ is analytic

or not

$$w = 2xy + i(x^2 - y^2)$$

$$u + iv = 2xy + i(x^2 - y^2)$$

$$u = 2xy$$

$$v = x^2 - y^2$$

$$u_x = 2y$$

$$v_x = 2x$$

$$u_x = -v_y$$

$$u_x = 2y \quad - \quad v_x = 2x$$

$$u_y = 2x \quad v_y = -2y$$

$$u_x \neq v_y \quad \text{CR are not satisfied}$$

$$v_x \neq -u_y$$

$\therefore$ The given fcn is not analytic.

⑤ Show $f(z) = \frac{1}{z}$ is analytic everywhere

Solution:-

$$f(z) = \frac{1}{z}$$

$$u + iv = \frac{1}{x + iy}$$

$$= \frac{1}{x + iy} \times \frac{x - iy}{x - iy}$$

$$u + iv = \frac{x - iy}{x^2 + y^2}$$

$$u + iv = \frac{x}{x^2 + y^2} + \frac{-iy}{x^2 + y^2}$$

$$u = \frac{x}{x^2 + y^2}$$

$$v = \frac{-y}{x^2 + y^2}$$

$$U_n = \frac{(x^2 + y^2) (1 - x(2n+1))}{(x^2 + y^2)^2}$$

$$V_x = \frac{y}{(x^2 + y^2)^2} \quad (2n+1)$$

$$V_x = \frac{2xy}{(x^2 + y^2)^2}$$

$$V_y = - \frac{[(x^2 + y^2) (1 - y(2n+1))]}{(x^2 + y^2)^2}$$

$$V_y = - \frac{[x^2 + y^2 - 2y^2]}{(x^2 + y^2)^2}$$

$$V_y = - \frac{[x^2 - y^2]}{(x^2 + y^2)^2}$$

$$V_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$u_x = V_y$$

$$V_x = -u_y$$

Hence given satisfies CR cond & analytic

$$U_n = \frac{x^2 + y^2}{(x^2 + y^2)^2}$$

$$U_n = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$V_y = \frac{-x}{(x^2 + y^2)^2} \quad (0,1)$$

$$u_y = \frac{-2xy}{(x^2 + y^2)^2}$$

Derive the analytic function

$$f(z) = z^n$$

$$= (e^{i\theta})^n$$

$$u + iv = z^n e^{in\theta}$$

$$= z^n (\cos n\theta + i \sin n\theta)$$

$$u + iv = z^n \cos n\theta + i z^n \sin n\theta$$

$$u = z^n \cos n\theta \quad v = z^n \sin n\theta$$

$$\frac{\partial u}{\partial z} = n z^{n-1} \cos n\theta \quad \frac{\partial v}{\partial z} = n z^{n-1} \sin n\theta$$

$$\frac{\partial u}{\partial \theta} = -n z^n \sin n\theta \quad \frac{\partial v}{\partial \theta} = n z^n \cos n\theta$$

$$\frac{\partial u}{\partial z} = \frac{1}{z} \frac{\partial v}{\partial \theta}$$

$$n z^{n-1} \cos n\theta = \frac{1}{z} n z^n \cos n\theta$$

$$n z^{n-1} \cos n\theta = n z^{n-1} \cos n\theta$$

$$\frac{\partial v}{\partial z} = -\frac{1}{z} \frac{\partial u}{\partial \theta}$$

$$n z^{n-1} \sin n\theta = -\frac{1}{z} (-n z^n \sin n\theta)$$

$$n z^{n-1} \sin n\theta = n z^{n-1} \sin n\theta$$

$\therefore$ given function z^n is an analytic function.

2. if $f(z) = (x-y)^2 + 2i(x+y)$ show that the CR equations are satisfied along the curve $x-y=1$

$$f(z) = (x-y)^2 + 2i(x+y)$$

$$u+iv = (x-y)^2 + i2(x+y)$$

$$u = (x-y)^2$$

$$v = 2(x+y)$$

$$u_x = 2(x-y)(1) \quad , \quad v_x = 2(1+0) = 2$$

$$u_y = 2(x-y)(-1) \quad , \quad v_y = 2(0+1) = 2$$

$$\text{Put } x-y = 1$$

$$u_x = 2 \quad , \quad v_x = 2$$

$$u_y = -2 \quad , \quad v_y = 2$$

$$u_x = v_y$$

$$v_x = -u_y$$

Construction of analytic function:-

(i) Milne - Thomson method:-

Type 1:

When the real part of

$f(z) = u$ is given then use

have to construct an analytic function
 $f(z) = \int [u_x(z,0) - i u_y(z,0)] dz + c$

Problems:-

if $u(x,y) = e^x (x \cos y - y \sin y)$
find $f(z)$ so that $f(z)$ is analytic

Solution:-

$$u = e^x x \cos y - e^x y \sin y$$

$$u_x = [e^x(1) + x e^x] \cos y - e^x y \sin y$$

$$u_x(z,0) = [e^x + x e^x] \cos 0 - e^x 0 \sin 0$$

$$= [e^x + x e^x] (1) = 0$$

$$u_x(z,0) = (1+x) e^x$$

$$u_y = -e^x x \sin y - e^x [y \cos y + \sin y(1)]$$

$$u_y(z,0) = 0$$

$$f(z) = \int [u_x(z,0) - i u_y(z,0)] dz + c.$$

$$f(z) = \int [(1+x) e^x - i(0)] dz + c.$$

$$f(z) = \int (1+x) e^x dz + c.$$

$$u = 1 + z, \quad dv = e^z dz$$

$$u' = 1 \quad v = e^z$$

$$u'' = 0 \quad v = e^z$$

$$\cos i\theta = \cosh \theta$$

$$\sin i\theta = i \sinh \theta$$

$$f(z) = (1+z)e^z - e^z + c$$

$$f(z) = e^z + ze^z - e^z + c$$

$$f(z) = ze^z + c$$

If $u = 2(\sin 2x)$ Find $w = f(z)$ such
 $e^{2y} + e^{-2y} - 2\cos 2x$

that $f(z)$ is Analytic. Also find v .

$$\cosh \theta = \frac{e^\theta + e^{-\theta}}{2}$$

$$u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x}$$

$$2 \cosh \theta = e^\theta + e^{-\theta}$$

$$u = \frac{2 \sin 2x}{2 \cosh 2y - 2\cos 2x}$$

$$u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$u_x = \frac{(\cosh 2y - \cos 2x) 2\cos 2x - \sin 2x (0 + 2\sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$$u_x(x,0) = \frac{(\cos^2 z - \cos 2z) 2 \cos 2z - \sin 2z (2 \sin 2z)}{(\cos 0 - \cos 2z)^2}$$

$$= \frac{2 \cos 2z - 2 \cos^2 2z - 2 \sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2 \cos 2z - 2 (\cos^2 2z + \sin^2 2z)}{(1 - \cos 2z)^2}$$

$$= \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2}$$

$$= \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2}$$

$$u_x(x,0) = \frac{-2}{1 - \cos 2z}$$

$$= \frac{-2}{2 \sin^2 z}$$

$$u_x(x,0) = -\operatorname{cosec}^2 z$$

$$u = \frac{\sin 2x}{\cos 2y - \cos 2z}$$

diff w.r.t. z

$$u_y = \frac{-\sin 2x}{(\cos hy - \cos 2x)^2} \cdot 2 \sinh 2y$$

$$u_y(z, 0) = 0$$

$$f(z) = \int [u_x(x, 0) - i u_y(x, 0)] dx + C$$

$$= \int [-\operatorname{cosec}^2 x - i(0)] dx + C$$

$$f(z) = \int \operatorname{cosec}^2 x dx + C$$

$$f(z) = \cot x + C$$

$$f(x) = \frac{\cos x}{\sin x}$$

$$u + iv = \frac{\cos(x + iy)}{\sin(x + iy)}$$

$$u + iv = \frac{2 \cos(x + iy) \times \sin(x - iy)}{2 \sin(x + iy)}$$

$$= \frac{\sin(2x) - \sin 2(2iy)}{\cos(2iy) - \cos(2x)}$$

$$= \frac{\sin 2x - i \sinh 2y}{\cosh 2y - \cos 2x}$$

$$f(z) = \cot z$$

$$f(z) = \frac{1}{\tan z}$$

$$u(x, y) = \frac{1}{\tan(x + iy)}$$

$$= \frac{\cos(x + iy)}{\sin(x + iy)}$$

$$\frac{\cos(x + iy)}{\sin(x + iy)}$$

$$\cot z$$

$$u + iV = \left(\frac{\sinh 2y}{\cosh 2y - \cos 2x} \right) + i \left(\frac{-\sinh 2x}{\cosh 2y - \cos 2x} \right)$$

$$V = \frac{-\sinh 2x}{\cosh 2y - \cos 2x}$$

Find $f(z) = u + iV$ for the real part is $\log(x^2 + y^2)$ also

find V .

solution:-

$$u = \log(x^2 + y^2)$$

$$u_x = \frac{1}{x^2 + y^2} \cdot 2x$$

$$u_x(x, y) = \frac{2x}{x^2 + y^2}$$

$$u_x(x, y) = \frac{2}{x}$$

$$u_y = \frac{1}{x^2 + y^2} \cdot 2y$$

$$u_y(x, y) = 0$$

$$\begin{aligned} \log(a + ib) &= \\ \log \sqrt{a^2 + b^2} + i \tan^{-1} \left(\frac{b}{a} \right) \end{aligned}$$

$$f(z) = \int [u_x(x,0) - i u_y(x,0)] dz + C$$

$$= \int \left[\frac{2}{z} - i(0) \right] dz + C$$

$$= 2 \int \frac{1}{z} dz + C$$

$$f(z) = 2 \log z + C$$

$$u + iv = 2 \log(x + iy) + C$$

$$u + iv = 2 \left[\log \sqrt{x^2 + y^2} + i \tan^{-1}(y/x) \right]$$

$$u + iv = 2 \left[\frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}(y/x) \right]$$

$$u + iv = \log(x^2 + y^2) + i 2 \tan^{-1}(y/x)$$

$$v = 2 \tan^{-1}(y/x)$$

find $f(z) = u + iv$ for

$$u = \frac{x}{x^2 + y^2}$$

real part:

$$u = \frac{x}{x^2 + y^2}$$

$$u_x = \frac{(x^2+y^2)(1) - (x)(2x)}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$$

$$u_x(x,0) = \frac{(x)^2 + (0) - 2(x)^2}{x^2 + 0^2}$$

$$= \frac{x^2 - 2x^2}{x^4}$$

$$= \frac{\cancel{x^2} (1-2)}{x^4}$$

$$u_x(x,0) = \frac{(1-2)}{x^2} = \frac{-1}{x^2}$$

$$u_y = \frac{(x^2+y^2)(0) - (x)(2y)}{(x^2+y^2)^2}$$

$$= \frac{-2xy}{(x^2+y^2)^2}$$

$$u_y(x,0) = \frac{-2(x)(0)}{(x^2+0^2)^2}$$

$$u_y(x,0) = 0$$

$$f(z) = \int [u_x(z,0) - i u_y(z,0)] dz + C$$

$$= \int \left(-\frac{1}{z^2} - i(0) \right) dz + C$$

$$= - \int \frac{1}{z^2} dz + C$$

$$= - \int z^{-2} dz + C$$

$$= - \int \frac{z^{-2+1}}{-2+1} + C$$

$$= - \frac{z^{-1}}{-1} + C$$

$$f(z) = \frac{1}{z} + C$$

Type II:

when the imaginary part of v is ~~not~~ given then the analytic function

$$f(z) = \int [v_y(z,0) + i v_x(z,0)] dz + C.$$

① Find an analytic function whose imaginary part is $3x^2y - y^3$

$$v = 3x^2y - y^3$$

$$v_x = 6xy$$

$$v_y = 3x^2 - 3y^2$$

$$u_x(x, 0) = 0$$

$$v_y(x, 0) = 3x^2$$

$$f(z) = \int [v_y(x, 0) + i v_x(x, 0)] dz + C$$

$$f(z) = \int [3x^2 + i(0)] dz + C$$

$$= \frac{3x^3}{3} + C$$

$$f(z) = z^3 + C$$

② Find analytic function whose imaginary part is $v = \frac{2 \sin x \sinh y}{\cos 2x + \cosh 2y}$

$$u_x = \frac{(\cos 2x + \cosh 2y) 2 \cos x \sinh y - 2 \sin x \sinh y (-2 \sin 2x + 0)}{(\cos 2x + \cosh 2y)^2}$$

$$u_x(x, 0) = 0$$

$$v_y = \frac{(\cos 2x + \cosh 2y) \{ \sin x \cosh y - i \sin x \sinh y \}}{(\cos 2x + \cosh 2y)}$$

$$u_y(z, 0) = \frac{(\cos 2z + \cos 0) \{ \sin z \cos 0 \}}{(\cos 2z + \cos 0)^2}$$

$$= \frac{(\cancel{\cos 2z} + 1) \{ \sin z \}}{(\cancel{\cos 2z} + 1)^2}$$

$$= \frac{\sin z}{\cos 2z + 1}$$

$$= \frac{\sin z}{2 \cos^2 z}$$

$$f(z) = \int [v_y(z, 0) + i v_x(z, 0)] dz + C$$

$$= \int \left[\frac{\sin z}{2 \cos^2 z} + i(0) \right] dz + C$$

$$d(\sec z) = \sec z \tan z$$

$$\int d(\sec z) = \int \sec z \tan z$$

$$\sec z = \int \sec z \tan z$$

$$f(z) = \frac{1}{2} \int \frac{\sin z}{\cos^2 z} dz + C$$

$$= \frac{1}{2} \int \frac{\sin z}{\cos z} \cdot \frac{1}{\cos z} dz + C$$

$$= \frac{1}{2} \int (\tan x \sec x) dx + C$$

$$= \frac{1}{2} \int \sec x + C$$

$$f(z) = \frac{1}{2} \boxed{\sec x + C}$$

Type III:-

Combination of real part & imaginary part

Procedure:-

$$f(z) = u + iv \quad \text{--- (1)}$$

$$if(z) = iu - v \quad \text{--- (2)}$$

multiply

multiply

eqn (1)

eqn (2)

by co-efficient of

by co-efficient of u

Adding

co-efficient of v and u.

$$\textcircled{1} \rightarrow uv + i v^2$$

$$\textcircled{2} \rightarrow i u^2 - v^2$$

$$f(z) =$$

Find the analytic function of $f(z)$ given that

$$u + v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$f(z) = u + iv$$

$$if(z) = iu - v$$

$$f(z) + if(z) = (u + iv) + (iu - v)$$

$$(1+i)f(z) = (u - v) + i(u + v)$$

$$F(z) = U + iV$$

$$F(z) = (1+i)f(z)$$

$$U = u - v$$

$$V = u + v \quad \text{imaginary part constant}$$

$$V = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$V_n = \frac{(\cosh 2y - \cos 2x) 2 \cos 2x - \sin 2x (0 + 2 \sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$$V_n(z, i0) = \frac{(\cosh 0 - \cos 2z) 2 \cos 2z - \sin 2z (2 \sin 2z)}{(\cosh 0 - \cos 2z)^2}$$

$$= \frac{2\cos 2z - 2\cos^2 2z - 2\sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2\cos 2z - 2(\cos^2 2z + \sin^2 2z)}{(1 - \cos 2z)^2}$$

$$= \frac{2\cos 2z - 2}{(1 - \cos 2z)^2}$$

$$= \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2}$$

$$= \frac{-2}{1 - \cos 2z} = \frac{-2}{2\sin^2 z}$$

$$V_x(z, 0) = -\operatorname{cosec}^2 z$$

$$V_y = \frac{-\sinh 2x}{(\cosh 2y - \cosh 2x)^2} \quad \& \sinh 2y$$

$$V_y(z, 0) = 0$$

$$F(z) = \int [v_y(z,0) + i v_x(z,0)] dz + c$$

$$F(z) = \int [0 + i(-\cos e^{2z})] dz + c$$

$$F(z) = -i \int \cos e^{2z} dz + c$$

Integration

$$F(z) = i \cot z + c$$

$$(1+i) f(z) = i \cot z + c$$

$$f(z) = \frac{i \cot z}{1+i} + \frac{c}{1+i} \quad (\text{Conjugate})$$

$$f(z) = \frac{i \cot z}{1+i} \times \frac{1-i}{1-i} + c_1$$

$$f(z) = \frac{(i - i^2) \cot z}{(1)^2 - (i)^2} + c_1$$

$$f(z) = \frac{(i+1) \cot z}{2} + c_1$$

Find the analytic function $f(z)$ given that $u - v = e^x (\cos y - \sin y)$

$$u - v = e^x (\cos y - \sin y)$$

$$f(z) + i f(z) = (u + iv) + i(u - v)$$

$$(1+i) f(z) = (u - v) + i(u + v)$$

$$F(z) = u + iV$$

$$f(z) = (1+i)g(z)$$

$$u = u - v$$

$$v = u + v$$

$$v = e^x (\cos y - \sin y)$$

$$u_x = e^x (\cos y - \sin y)$$

$$u_x(z,0) = e^x (\cos 0 - \sin 0)$$

$$u_x(x,0) = e^x$$

$$u_y = e^x (-\sin y - \cos y)$$

$$u_y(z,0) = e^x (-\sin 0 - \cos 0)$$

$$u_y(x,0) = e^x (0-1)$$

$$u_y(z,0) = -e^x$$

$$F(z) = \int [u_x(z,0) - i u_y(z,0)] dx + c$$

$$F(z) = \int [e^x - i(-e^x)] dx + c$$

$$F(z) = \int (e^x + i e^x) dx + c$$

$$F(z) = \int (1+i) e^x dx + c$$

$$F(z) = (1+i) e^x + c$$

1. Inverse (or) linear transformation
 $(1+i) f(z) = (1+i) e^z + C$
 $f(z) = \frac{(1+i) e^z}{1+i} + \frac{C}{1+i}$
 Property of analytic function
 $f(z) = e^z + C_1$

(2) Find the analytic function given that

$$2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$f(z) = u + i v$$

$$i f(z) = i u - v$$

$$3 f(z) = 3u + i 3v$$

$$2 i f(z) = i 2u - 2v$$

$$3 f(z) + 2 i f(z) = (3u + i 3v) + (i 2u - 2v)$$

$$(3 + 2i) f(z) = (3u - 2v) + i (2u + 3v)$$

$$f(z) = U + i V$$

$$f(z) = (3 + 2i) f(z)$$

$$U = 3u - 2v$$

$$V = 2u + 3v$$

for previous problem question
 $u + v$

$$V = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$V_x(z,0) = -\operatorname{cosec}^2 x$$

$$V_y(z,0) = 0$$

$$F(z) = \int [V_y(z,0) + iV_x(z,0)] dz + C$$

$$F(z) = \int [0 + i(-\operatorname{cosec}^2 x)] dx + C$$

$$F(z) = -i \int \operatorname{cosec}^2 x dx + C$$

$$F(z) = i \cot x + C$$

$$(3+2i)f(z) = i \cot x + C$$

$$f(z) = \frac{i \cot x}{3+2i} + \frac{C}{3+2i}$$

$$f(z) = \frac{i \cot x}{3+2i} \times \frac{3-2i}{3-2i} + C_1$$

$$f(z) = \frac{(3i+2) \cot x}{13} + C_1$$

Properties of harmonic function:-

Property 1:-

show that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$

Proof:-

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i} = \frac{-i}{2}(z - \bar{z})$$

$$\frac{\partial x}{\partial z} = \frac{1}{2}$$

$$\frac{\partial y}{\partial z} = \frac{-i}{2}$$

$$\frac{\partial x}{\partial \bar{z}} = \frac{1}{2}$$

$$\frac{\partial y}{\partial \bar{z}} = \frac{i}{2}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x} \cdot \frac{\partial x}{\partial z} + \frac{\partial}{\partial y} \cdot \frac{\partial y}{\partial z}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x} \cdot \frac{1}{2} + \frac{\partial}{\partial y} \left(\frac{-i}{2} \right)$$

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \rightarrow \text{①}$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial}{\partial y} \frac{\partial y}{\partial \bar{z}}$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial}{\partial y} \left(\frac{1}{2} \right)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \rightarrow (2)$$

① × ②

$$\frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

$$\frac{\partial^2}{\partial z \partial \bar{z}} = \frac{1}{4} \left(\left(\frac{\partial}{\partial x} \right)^2 - \left(i \frac{\partial}{\partial y} \right)^2 \right)$$

$$4 \frac{\partial^2}{\partial z \partial \bar{z}} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \quad /$$

Property 2:-
 if $f(z)$ is a regular function of z
 then prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$

17)

Consider :-

all in 10 min u can

→ give proof (or) 6m

wer

proof:

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \left(|f(z)|^2 \right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2$$

$$= 4 |f(z)|^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$$

$$|z|^2 = z \bar{z}$$

$$= \frac{4 \partial^2}{\partial z \partial \bar{z}} |f(z)|^2$$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}} f(z) f(\bar{z})$$

$$= 4 \frac{\partial}{\partial z} \cdot \frac{\partial}{\partial \bar{z}} f(z) f(\bar{z})$$

$$= 4 \frac{\partial}{\partial z} f(z) f'(\bar{z})$$

$$= 4 \cdot f'(z) f'(\bar{z})$$

$$= 4 |f'(z)|^2$$

$$= \text{RHS}$$

Property 3:-

if $f(z) = u + iv$ is analytic prove

$$\text{that } \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log |f'(z)| = 0$$

$$LHS = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log |f'(z)|$$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \log |f'(z)|$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \log |f'(z)|$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \log |f'(z)|^{2/2}$$

$$= 4 \frac{\partial}{\partial z} \cdot \frac{\partial}{\partial \bar{z}} \cdot \frac{1}{2} \log |f'(z)|^2$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \frac{1}{2} \log (f'(z) f'(\bar{z}))$$

$$= 2 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \left[\log f'(z) + \log f'(\bar{z}) \right]$$

$$= 2 \frac{\partial}{\partial z} \left[0 + \frac{1}{f'(\bar{z})} f''(\bar{z}) \right]$$

$$= 0$$

$$= RHS$$

Property 4:- $f(z) = u + iv$ is analytic

prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log |f(z)| = 0$

LHS:-

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log |f(z)|$$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \log |f(z)|$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \log |f^2(z)|^{1/2}$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \frac{1}{2} \log |f^2(z)|$$

$$= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \frac{1}{2} \log (f(z) f(\bar{z}))$$

$$= 2 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} \left[\log f(z) + \log f(\bar{z}) \right]$$

$$= 2 \frac{\partial}{\partial z} \left[0 + \frac{1}{f(\bar{z})} f'(\bar{z}) \right]$$

$$= 0$$

$$= \text{RHS}$$

Problems on harmonic functions:-

Prove that $u = e^x \cos y$ is a harmonic function.

$$u = e^x \cos y$$

$$u_x = e^x \cos y$$

$$u_{xx} = e^x \cos y$$

$$u_y = -e^x \sin y$$

$$u_{yy} = -e^x \cos y$$

$$u_{xx} + u_{yy} = e^x \cos y - e^x \cos y$$

$$u_{xx} + u_{yy} = 0$$

Show that the function $u = 2xy + 3y$ is harmonic

$$u = 2xy + 3y$$

$$u_x = 2y$$

$$u_y = 2x + 3$$

$$u_{xx} = 0$$

$$u_{yy} = 0$$

$$u_{xx} + u_{yy} = 0 + 0 = 0$$

u satisfies harmonic equation

Prove that the function $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic also find conjugate harmonic function v and corresponding harmonic function $f(z)$

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = 3x^2 - 3y^2 + 6x$$

$$u_x(z, 0) = 3x^2 + 6x$$

$$u_{xx} = 6x + 6$$

$$u_y = -6xy - 6y$$

$$u_y(z, 0) = 0$$

$$u_{yy} = -6x - 6$$

$$u_{xx} + u_{yy} = 6x + 6 - 6x - 6$$

$$u_{xx} + u_{yy} = 0$$

u satisfies Laplace equation

$\therefore u$ is harmonic function.

$$f(z) = \int [u_x(z, 0) - i u_y(z, 0)] dz + C$$

$$= \int [(3x^2 + 6x) - i(0)] dz + C$$

$$f(z) = \int (3z^2 + 6z) dz + C$$

$$f(z) = \frac{3z^3}{3} + \frac{6z^2}{2} + C$$

$$f(z) = z^3 + 3z^2 + C$$

$$f(z) = (z+3)z^2 + C$$

$$u+iv = (x+iy+3)(x+iy)^2 + C$$

$$u+iv = (x+iy+3)(x^2 + i2xy - y^2)$$

$$u+iv = x^3 + i2x^2y - xy^2 + i x^2y - 2xy^2 - iy^3$$

$$+ 3x^2 + i6xy - 3y^2$$

$$u+iv = (x^3 - 3xy^2 + 3x^2 - 3y^2) + i(3x^2y - y^3 + 6xy)$$

$$v = 3x^2y - y^3 + 6xy$$

Verify that the fn $u = e^x(\cos y - \sin y)$ is harmonic (or) not. If it is harmonic find the conjugate harmonic v . So $f(z) = u+iv$ is analytic.

$$u = e^x(\cos y - \sin y)$$

$$u_x = e^x(\cos y - \sin y)$$

$$u_{xx} = e^x(\cos y - \sin y)$$

$$e^x \cos y$$

$$u_y = -e^x \sin y - e^x \cos y$$

$$f(z) = \int e^z dz = e^z + C = e^x(\cos y - \sin y) + i(e^x \sin y + e^x \cos y) = e^x(\cos y - \sin y) + i e^x(\sin y + \cos y)$$

Show that $v = e^x (x \cos y - y \sin y)$ is harmonic also find the analytic function $f(z)$.

$$v = e^x (x \cos y - y \sin y)$$

$$v = e^x x \cos y - e^x y \sin y$$

$$v_x = [e^x + x e^x] \cos y - e^x y \sin y$$

$$v_x(z, 0) = e^x + x e^x = (1+x) e^x$$

$$v_{xx} = (e^x + x e^x + e^x) \cos y - e^x y \sin y$$

$$v_{xx} = 2 e^x \cos y + x e^x \cos y - e^x y \sin y$$

$$v_y = -e^x x \sin y - e^x [y \cos y + \sin y]$$

$$v_y(z, 0) = 0$$

$$v_{yy} = -e^x x \cos y - e^x [-y \sin y + \cos y + \cos y]$$

$$v_{yy} = -e^x x \cos y + e^x y \sin y - 2 \cos y e^x$$

$$v_{xx} + v_{yy} = 2 e^x \cos y + x e^x \cos y - e^x y \sin y - e^x x \cos y + e^x y \sin y - 2 \cos y e^x$$

$$v_{xx} + v_{yy} = 0$$

$$f(z) = \int [v_y(z,0) + i v_x(z,0)] dz + C$$

$$= \int [0 + i(1+z)e^z] dz + C$$

$$f(z) = i \int (1+z)e^z dz + C$$

Using Bernoulli's equation

$$u = 1+z \quad , \quad dv = e^z dz$$

$$u' = 1 \quad , \quad v = e^z$$

$$u'' = 0 \quad , \quad v_1 = e^z$$

$$f(z) = i [(1+z)e^z - e^z] + C$$

$$f(z) = i [e^z + ze^z - e^z] + C$$

$$f(z) = i ze^z + C$$

Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic and determine its conjugate also find $f(z)$.

$$u = \frac{1}{2} \log(x^2 + y^2)$$

$$u_x = \frac{1}{2} \frac{1}{x^2 + y^2} (2x)$$

$$u_x = \frac{x}{x^2 + y^2}$$

$$u_x(x,0) = \frac{x}{x^2+0} = \frac{1}{x}$$

$$u_{xx} = \frac{x^2+y^2(1) - x(2x)}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2 - 2x^2}{(x^2+y^2)^2}$$

$$= \frac{y^2 - x^2}{(x^2+y^2)^2}$$

$$u_y = \frac{1}{2} \frac{1}{x^2+y^2} \quad (2y)$$

$$u_y(x,0) = 0$$

$$u_{yy} = \frac{(x^2+y^2)(1) - y(2y)}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2 - 2y^2}{(x^2+y^2)^2}$$

$$= \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$u_{xx} + u_{yy} = \frac{y^2 - x^2}{(x^2+y^2)^2} + \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$u_{xx} + u_{yy} = 0$$

$$f(z) = \int [u_x(z,0) - i u_y(z,0)] dz + C$$

$$= \int \left[\frac{1}{z} - i(0) \right] dz + C$$

$$f(z) = \int \frac{1}{z} dz + C$$

$$f(z) = \log z + C$$

$$u + iv = \log(x + iy) + C$$

$$u + iv = \log \sqrt{x^2 + y^2} + i \tan^{-1}(y/x) + C$$

$$v = \tan^{-1}(y/x)$$

Find the constant 'a' so that $u = ax^2 - y^2 + xy$ is harmonic find an analytic function $f(z)$ for which u is real part.

given u is harmonic $\therefore$ it satisfies Laplace equation $u_{xx} + u_{yy} = 0$

$$u = ax^2 - y^2 + xy$$

$$u_x = 2x + y$$

$$v_x = 2y$$

$$u_y = -2y$$

$$v_y = -1$$

$$u_{yy} = -2$$

$$v_{yy} = 0$$

$$u_{xx} + u_{yy} = 0$$

$$2a - 2 = 0$$

$$2(a-1) = 0$$

$$a-1 = 0$$

$$a = 1$$

$$u = x^2 - y^2 + xy$$

$$v_x = 2y$$

$$da = 2x + y$$

$$v_y = 2x - 1$$

$$u_x(x, 0) = 2x$$

$$u_y = -2y + x$$

$$u_y(x, 0) = x$$

$$f(z) = \int [u_x(x, 0) - i u_y(x, 0)] dx + C$$

$$f(z) = \int (2x - ix) dx + C$$

$$f(z) = \int [(2-i)x] dx + C$$

$$= (2-i) \int x dx + C = \frac{(2-i) \cdot x^2}{2} + C$$